Example 12.12

Write the statements in words corresponding to $\neg p$, $p \wedge q$, $p \vee q$ and $q \vee \neg p$, where p is 'It is cold' and q is 'It is raining.'

Solution

(1) $\neg p$: It is not cold.

(2) $p \wedge q$: It is cold and raining.

(3) $p \vee q$: It is cold or raining.

(4) $q \vee \neg p$: It is raining or it is not cold

Observe that the statement formula $\neg p$ has only 1 variable p and its truth table has $2 = (2^1)$ rows. Each of the statement formulae $p \wedge q$ and $p \vee q$ has two variables p and q. The truth table corresponding to each of them has $4 = (2^2)$ rows. In general, it follows that if a statement formula involves n variables, then its truth table will contain 2^n rows.

Example 12.13

How many rows are needed for following statement formulae?

(i) $p \vee \neg t \wedge (p \vee \neg s)$ (ii) $((p \wedge q) \vee (\neg r \vee \neg s)) \wedge (\neg t \wedge v)$

Solution

(i) $(p \vee \neg t) \wedge (p \vee \neg s)$ contains 3 variables p, s ,and t. Hence the corresponding truth table will contain $2^3 = 8$ rows.

(ii) $((p \wedge q) \vee (\neg r \vee \neg s)) \wedge (\neg t \wedge v)$ contains 6 variables p, q, r, s, t, and v. Hence the corresponding truth table will contain $2^6 = 64$ rows.

Conditional Statement

Definition 12.13

The conditional statement of any two statements p and q is the statement, "If p, then q" and it is denoted by $p \rightarrow q$. Here p is called the **hypothesis** or **antecedent** and q is called the **conclusion** or **consequence**. $p \rightarrow q$ is false only if p is true and q is false. Otherwise it is true.

Truth table for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Table 12.7

Example 12.14

Consider $p \rightarrow q$: If today is Monday, then 4 + 4 = 8.

Here the component statements p and q are given by,

p: Today is Monday; q: 4 + 4 = 8.

The truth value of $p \rightarrow q$ is T because the conclusion q is T.

An important point is that $p \rightarrow q$ should not be treated by actually considering the meanings of p and q in English. Also it is not necessary that p should be related to q at all.

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